

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built **multi-agent system** via **OpenAI Agents SDK** + **MCP** tool-use handling **millions of daily transactions** at **99%+ uptime**.
- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.
- Built **evaluation harnesses** measuring LLM output quality, safety, **hallucination rates**, and grounding across agent versions; iterated on **eval results** and production usage data.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** migrating synchronous serving to async **vLLM** on **GCP Cloud Run** with autoscaling.
- Improved **RAG retrieval recall 61% to 89%** on a 2K-query eval set by rebuilding chunking with **hybrid dense-sparse retrieval** and **re-ranking**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and **SQL-backed distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized failure-handling contracts across **25+ financial services**, reducing production escalations **60%** in one quarter via consistent **retry, timeout, and propagation** semantics.
- Cut nightly financial data aggregation blocking downstream **settlement** from **8 minutes to 90 seconds** by redesigning **SQL** with **indexed views** and **batch partitioning**.

PROJECTS

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP-compatible server**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, TypeScript, Go, Rust, C++, Java, SQL, HTML/CSS

Agents: OpenAI Agents SDK, LangGraph, MCP, Tool Use, Agent Failure Recovery, Evals

Systems: Distributed Executors, Checkpointing, Docker, AWS Lambda, Kubernetes

Data: PostgreSQL, Redis, S3, Kafka

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer